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Team members:

Person code	Surname	Name	Bachelor (type and university)
10770766	Cavagna	Francesco	Ingegneria aerospaziale (Politecnico di Milano)
10814647	Corazza	Andrea	Ingegneria aerospaziale (Politecnico di Milano)
11093867	Pezzi	Francesco	Ingegneria aerospaziale (Università degli Studi di Roma "La Sapienza")
10804340	Rizzi	Giovanni Federico	Ingegneria aerospaziale (Politecnico di Torino)

Project specifications:

	Assigned specification	Modifications	Motivation for modification
Platform	minisat (100-500 kg)		
Attitude parameters	direction cosines		
Mandatory sensor	Sun sensor	a) Magnetometer b) 2 Sun sensors	a) two measurements are required for attitude determination b) improve attitude determination performance
Actuators	3 magnetic coils 1 reaction wheel		

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1 Introduction

The present report describes the design of the attitude dynamics control model of a minisat. It is considered to be on a Sun-synchronous orbit and it is required to point towards the Sun. The satellite has a mass of approximately 130 kilograms and it is equipped with a Sun sensor, a magnetometer, three magnetic coils and a reaction wheel. In order to achieve a realistic representation of the behaviour of the satellite the simulation takes into account the most influential disturbance torques produced by the surrounding environment and the uncertainties and errors linked to the performances of the selected ADCS components.

1.1 Satellite configuration

The minisat configuration draws inspiration from the ESA spacecraft Proba-2, whose characteristics are quite similar to those prescribed in the present essay. It is assumed that the satellite is composed of a central body ($0.6\text{ m} \times 0.8\text{ m} \times 0.6\text{ m}$) and two deployable panels ($0.6\text{ m} \times 0.8\text{ m} \times 0.015\text{ m}$). The mass of the spacecraft is distributed in such a way that the centre of mass coincides with the geometrical centre of the main body.

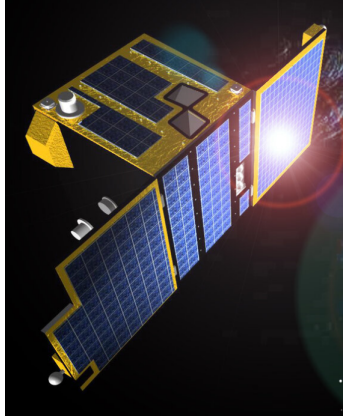


Figure 1: Proba-2 spacecraft

The inertia matrix of the satellite with the deployed solar panels is:

$$\underline{\underline{\mathbf{I}}} = \begin{bmatrix} 11.5536 & 0 & 0 \\ 0 & 12.8402 & 0 \\ 0 & 0 & 15.1533 \end{bmatrix} \cdot \text{kg} \cdot \text{m}^2 \quad (1)$$

1.2 Orbit characterization

The Sun-synchronous orbit of the spacecraft is characterized by the parameters reported in Table 1:

Parameter	Value
a	7081 km
e	0
i	98.23°
Ω	0°
ω	0°

Table 1: Orbital parameters

where a is the semi-major axis, e is the eccentricity, i is the inclination over Earth's equatorial plane, Ω is the right ascension of the ascending node and ω is the argument of the perigee.

1.3 Sensors

The satellite is equipped with two Sun sensors and a magnetometer whose technical data are reported in the following tables:

1.3.1 Sun sensor - Aquila D02 [NewSpace Systems]

Accuracy	FOV	Sample frequency	SFE	SFA	D	D _{rw}
$\leq 0.2^\circ$	140°	5 Hz	625	500	0.01	0.01

Table 2: Sun sensor specifications [1]

1.3.2 Magnetometer - Pegasus-1

Accuracy	Measurement range	Sample frequency	SFE	SFA	D	D _{rw}
$\leq 8 \text{ nT}$	$\pm 60,000 \text{ nT}$	$\leq 25 \text{ Hz}$	200	150	0.0001	0.0001

Table 3: Magnetometer specifications [2]

1.4 Actuators

The satellite is equipped with three magnetic coils and a reaction wheel. The magnetic coils are positioned along the principal inertia axes, and the reaction wheel is mounted in correspondence of the z axis of the spacecraft. As a reasonable limit, we imposed to every actuator a maximum torque of $2Nm$ and a minimum one of $-2Nm$. A misalignment was also introduced, modelled as a product by a random value, constant throughout each simulation.

2 Model description

2.1 Attitude parameters

The attitude dynamics of the minisat is governed by Euler's equation [4]

$$\mathbf{I} \dot{\underline{\omega}} = \mathbf{I} \underline{\omega} \times \underline{\omega} + \underline{\mathbf{M}}_D + \underline{\mathbf{M}}_C \quad (2)$$

$\underline{\omega}$ is the angular velocity of the spacecraft, $\underline{\mathbf{M}}_D$ is the disturbing torque and $\underline{\mathbf{M}}_C$ is the control torque.

This equation is integrated numerically in order to calculate $\underline{\omega}$ which is necessary to compute the attitude kinematics of the spacecraft. In fact, it can be demonstrated that the time derivative of the direction cosine matrix $\mathbf{A}$ is related to the angular velocity $\underline{\omega}$ through the following equation [4]:

$$\frac{d\mathbf{A}}{dt} = -[\underline{\omega} \wedge] \mathbf{A}(t) \quad (3)$$

where $[\underline{\omega} \wedge]$ is a skew matrix built from the components of the vector $\underline{\omega}$. Once $d\mathbf{A}/dt$ is known it can be integrated in order to update the matrix $\mathbf{A}$. However, since integration is carried out numerically, the orthogonality of the direction cosine matrix is lost after some steps, therefore it is necessary to apply an orthogonalization process as the one described by the following equation [4]:

$$\mathbf{A}_{k+1}(t) = \mathbf{A}_k(t)^* 3/2 - \mathbf{A}_k(t)^* \mathbf{A}_k^T(t)^* \mathbf{A}_k(t)/2 \quad (4)$$

2.2 Disturbing torques

The disturbing torques considered in this project are the gravity gradient torque ($\underline{\mathbf{M}}_{GG}$), the Earth's magnetic field torque ($\underline{\mathbf{M}}_{MAG}$) and the solar radiation pressure torque ($\underline{\mathbf{M}}_{SPR}$). The sum of these three terms is the total disturbing torque ($\underline{\mathbf{M}}_D$). The aerodynamic torque is considered to be negligible since the altitude of the orbit is larger than 700 km. [4]

In order to compute these contributions it is necessary to know the position of the spacecraft along its orbit. The true anomaly of the satellite can be integrated from

$$\dot{\theta} = \frac{n(1 + e \cos \theta)^2}{(1 - e^2)^{3/2}} \quad (5)$$

where n is the average rotational rate of the orbit.

2.2.1 Gravity gradient torque

The gravity gradient torque is due to the non-uniformity of Earth's gravity field and it can be computed as follows [4]:

$$\underline{\mathbf{M}}_{GG} = \frac{3Gm_t}{R^3} \begin{Bmatrix} (I_z - I_y)c_2c_3 \\ (I_x - I_z)c_1c_3 \\ (I_y - I_x)c_1c_2 \end{Bmatrix} \quad (6)$$

where m_t is the mass of the Earth, R is the distance from the centre of the Earth to spacecraft and c_1, c_2, c_3 are the direction cosines of the radial direction from the centre of the Earth to the spacecraft in the body frame (i.e. the principal inertia axes frame).

2.2.2 Magnetic field torque

The disturbing torque produced by the Earth's magnetic field can be determined thanks to a vectorial equation [4]:

$$\underline{M}_{MAG} = \underline{j}_B \times \underline{b}_B \quad (7)$$

where $\underline{j}_B$ is the residual magnetic induction due to parasitic currents in the satellite and $\underline{b}_B$ is the Earth's magnetic field and they are both represented in the body frame of the spacecraft.

The Earth's magnetic field can be modeled as the gradient of a scalar function V , which is a series expansion of spherical harmonics [4]

$$V(r, \phi, \theta) = R \sum_{n=1}^k \left(\frac{R}{r} \right)^{n+1} \sum_{m=0}^n [g_n^m \cos(m\phi) + h_n^m \sin(m\phi)] P_n^m(\cos \theta) \quad (8)$$

R is the Earth's equatorial radius, r, ϕ and θ are the spherical coordinates of the position of the satellite and g_n^m and h_n^m are known as Gaussian coefficients and they are available up to order 13, which is the one chosen to simulate the real magnetic field. Once the magnetic field vector has been determined from the scalar potential V , it is converted in body frame.

The residual magnetic induction in the satellite $\underline{j}_B$ is assumed to be

$$\underline{j}_B = [0.03 \quad 0.15 \quad 0.03]^T A \cdot m^2 \quad (9)$$

Since there are no sufficient data to estimate the actual residual magnetic induction of the spacecraft, data reported on course slides was used, after multiplying it by a factor of 3.

2.2.3 Solar radiation pressure torque

The solar radiation pressure acting on the outer surface of the satellite produces a torque around the center of mass of the satellite itself. The SPR force for each of the satellite exposed faces is determined as:

$$\underline{F}_i = -PA_i \left(\hat{\underline{S}}_B \cdot \hat{\underline{N}}_{Bi} \right) \left[(1 - \rho_s) \hat{\underline{S}}_B + \left(2\rho_s \left(\hat{\underline{S}}_B \cdot \hat{\underline{N}}_{Bi} \right) + \frac{2}{3}\rho_d \right) \hat{\underline{N}}_{Bi} \right] \quad i = 1, \dots, n \quad (10)$$

P is the average pressure due to radiation and it can be estimated as $P = F_e/c$, where F_e is the power per unit surface (assumed to have a constant value of $1,358 \text{ W/m}^2$) and c is the speed of light. $\hat{\underline{N}}_{Bi}$ is the unit vector orthogonal to the surface in body frame, while $\hat{\underline{S}}_B$ is the unit vector from

the Sun to the satellite in body frame. Each surface of the satellite is characterized by a specular reflection coefficient (ρ_s) and a diffuse reflection coefficient (ρ_d). Their values for each body can be found in the file *satellitemodel*. The Sun vector is computed by rotating in the body frame $\hat{\underline{S}}_N$ (Sun vector in the inertial frame):

$$\hat{\underline{S}}_B = A_{B/N} \hat{\underline{S}}_N = A_{B/N} r_\odot \begin{Bmatrix} \cos(n_\odot t) \\ \cos(n_\odot t) \cos \varepsilon \\ \cos(n_\odot t) \sin \varepsilon \end{Bmatrix} \quad (11)$$

where $r_\odot$ is the average distance between the Earth and the Sun, $n_\odot$ is the mean angular velocity equal to $2\pi/T$ (T is the orbital period of the Earth around the Sun) and ε is the inclination of the equatorial plane over the ecliptic plane. The overall solar radiation pressure torque can be computed as [4]:

$$\underline{M}_{SPR} = \begin{cases} \sum_{i=1}^n \underline{r}_i \wedge \underline{F}_i & \hat{\underline{S}}_B \cdot \hat{\underline{N}}_{Bi} > 0 \\ 0 & \hat{\underline{S}}_B \cdot \hat{\underline{N}}_{Bi} < 0 \end{cases} \quad (12)$$

where $\underline{r}_i$ is the position of the centre of pressure of the flat surface with respect to the centre of mass. The condition on the sign of the dot product $\hat{\underline{S}}_B \cdot \hat{\underline{N}}_{Bi}$ is required in order to consider only the contributions of the surfaces which are actually illuminated.

2.3 Sensors

The models that simulate the Sun sensors and the magnetometer take into account the errors that affect the measurements of real sensors. The input for the sensors are respectively the Sun vector and the magnetic field vector (both in the body frame) as derived from the numerical solution of the equations. The input vector is normalized and then it is multiplied by two matrices: one of them takes into account the misalignment of the sensor while the other represents its bias. The misalignment matrix is built as

$$A_\epsilon = I - [\theta_\epsilon]^\wedge \quad (13)$$

where I is the identity matrix and $[\theta_\epsilon]^\wedge$ is the skew matrix built with the misalignment angle which is a small angle chosen randomly in this project, since it usually determined from empirical measurements. The bias matrix is built as the rotation matrix around a random direction and the rotation angle is determined according to literature sources [5] and [6]. The vector obtained after these two rotations is decomposed in its Cartesian components, and scale factor errors are taken into account for each one of them. Moreover, the errors due to white noise, bias instability and rate random walk are summed to each component of the vector. Once the "measured" vector has been calculated a quantizer, a saturation block and a zero-order hold are used to simulate the discrete nature of the sensor. The Sun sensor model includes also the limitations due to the FOV and the eclipse condition, and in case of failure of the sensor it is replaced by the numerical propagation of the Sun direction. For simplicity, the real direction of the Sun is considered, even though this leads to the paradox of the sensors working better when they are not actually functioning. Another

Sun sensor, whose observation axis is exactly mirrored with respect to the first one, was added in order to reduce the probability of the Sun being out of the FOV. The satellite is equipped with a magnetometer in addition to the Sun sensors because, as mentioned below, attitude determination requires at least two independent measurements.

2.4 Actuators

The satellite is equipped with two magnetic coils and one reaction wheel. The magnetic coils are positioned along the x and y axis of the spacecraft and the reaction wheel is on the z axis, so the control can be modeled as [4]:

$$\begin{pmatrix} M_x \\ M_y \\ M_z \end{pmatrix} = \begin{bmatrix} 0 & B_z & 0 \\ -B_z & 0 & 0 \\ B_y & -B_x & 1 \end{bmatrix} \begin{pmatrix} D_x \\ D_y \\ -\dot{h}_z \end{pmatrix} \quad (14)$$

where M_x , M_y and M_z are the components of the control moment $\underline{M}_c$, B_x , B_y and B_z are the components of the Earth's magnetic field, D_x and D_y are the magnitudes of the magnetic dipoles of the corresponding magnetic coils and $\dot{h}_z$ is the time derivative of the angular momentum of the reaction wheel.

The same formula was inverted to find the actuators commands (D_x, D_y and $-\dot{h}_z$):

The reaction wheel is characterized by a maximum angular velocity which is its saturation limit. When this threshold is reached no additional exchange of momentum may occur along the angular velocity direction and the wheel needs to be desaturated. In this project it was chosen to instantly reset the angular velocity when it reaches the threshold (i.e. the saturation limit), this decision was taken in order to simplify the desaturation mechanism.

$$\begin{pmatrix} D_x \\ D_y \\ -\dot{h}_z \end{pmatrix} = \left(\frac{1}{B_z} \right) \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ B_x & B_y & B_z \end{bmatrix} \begin{pmatrix} M_{cx} \\ M_{cy} \\ M_{cz} \end{pmatrix} \quad (15)$$

There might be a singularity when B_z gets closer to 0. In order to avoid this, a second actuators logic, involving the third magnetic coils, was implemented:

$$M_{\text{eff}} = \frac{1}{\|B\|^2} \cdot (B \wedge M_c) \wedge B \quad (16)$$

The actuators logic switches from the first to the second one when $B_z \leq 3 \cdot 10^{-8}$.

3 Mission requirements

The objective of the mission is to perform Nadir pointing, so the body frame is required to be aligned with the Local Vertical Local Horizontal frame (LVLH). For this reason the main performance parameter is $A_{B/L}$

$$A_{B/L} = A_{B/N} \cdot A_{N/L} \quad (17)$$

The desired attitude is characterized by :

- Desired attitude matrix: $A_{B/L} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
- Desired angular velocity: $\underline{\omega}_d = \{0, 0, n\}^T$
- Desired performance: $Pointingerror = \arccos\left(A_{B/L} \cdot \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}\right) \leq 0.5 \text{ deg}$

4 ADCS architecture

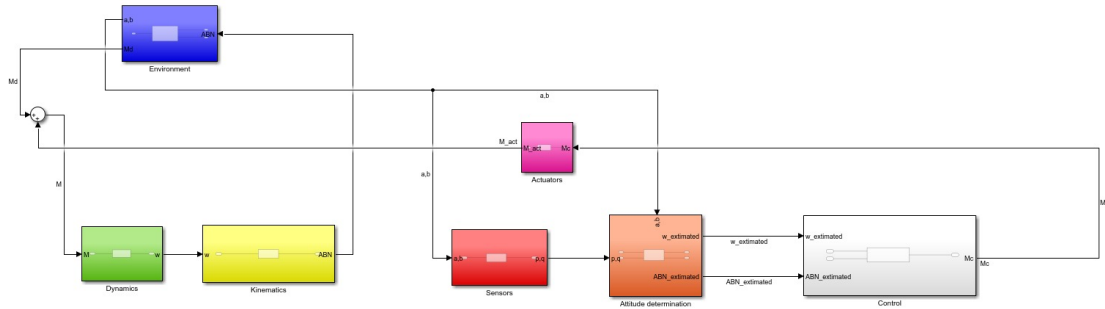


Figure 2: ADCS architecture

5 Attitude determination and control algorithms

5.1 Attitude determination algorithm

Attitude determination is carried out through an algorithm which requires at least two vector measurements (as the ones provided by the two sensors) and knowledge of the relative precision of the sensors. It is based on the minimization of a weighted error function [4]:

$$J = \frac{1}{2} \sum_{i=1}^N \alpha_i |\underline{s}_i - A \underline{v}_i|^2 \quad (18)$$

where s_i are the unit vectors derived from the measurements in body-fixed frame, and v_i are the corresponding unit vectors in the inertial frame. Since in the present project there are two available measurements, they can be used to build three unit vectors. If the measurements are denoted as p and r , and their corresponding directions in inertial frame are a and b , the unit vectors can be determined as follows [4]:

$$\begin{aligned} s_1 &= p \\ s_2 &= \frac{p \wedge q}{|p \wedge q|} \\ s_3 &= p \wedge s_2 \\ \\ v_1 &= a \\ v_2 &= \frac{a \wedge b}{|a \wedge b|} \\ v_3 &= a \wedge v_2 \end{aligned}$$

In particular, as p we chose the direction associated with the magnetic field, since our magnetometer was more accurate than the sun sensor, whose direction was linked to q .

In equation (18) α_i is a constant identifying the sensor precision, but in this case, since one of the directions is not derived from a real measurement, the corresponding weight factor is considered to be equal to the one related to the Sun sensor, since it is lower than the one associated to the magnetometer. It can be demonstrated that the optimal solution is the one that maximizes

$$\tilde{J} = \sum_{i=1}^N \alpha_i (\underline{s}_i^T A \underline{v}_i) = tr(AB^T) \quad (19)$$

This optimization problem can be translated into a vectorial equation

$$(S - \sigma I - I) \underline{g} + \underline{z} = \underline{0} \quad (20)$$

where $\underline{g}$ is the approximate Gibbs vector that provides the satellite attitude. S , σ and $\underline{z}$ are derived from the matrix B :

$$\begin{aligned}
B &= \sum_{i=1}^N \alpha_i s_i v_i^T \\
S &= B^T + B \\
\sigma &= \text{tr}(B) \\
\underline{z} &= [b_{23} - b_{32}, b_{31} - b_{13}, b_{12} - b_{21}]^T
\end{aligned}$$

From the approximate Gibbs vector, it's straightforward obtaining the Direction Cosine Matrix:

$$A = \frac{(1 - g^2)I + 2gg^T - 2[g\wedge]}{(1 + g^2)} \quad (21)$$

It's important to notice that directions a and b must be available on board, with approximated values, during a mission, and we modelled this for the magnetic direction, through an expansion of spherical harmonics up to 8th order available to the Attitude determination block, while the magnetic field is expanded up to the 13th order.

5.2 Angular velocity estimation algorithm

It is possible to exploit the determined DCM to find an estimation of the angular velocity w :

$$\dot{A} \approx -[\omega\wedge] \cdot A \quad (22)$$

$$[\omega\wedge] \approx -\dot{A} \cdot A^T \quad (23)$$

By computing the mean of the associated extra-diagonal values of $[\omega\wedge]$, it's possible to approximate w_x, w_y and w_z

5.3 Control

Our mission is divided into 3 phases: de-tumbling, slew manoeuvre and nadir pointing. We used a closed-loop control system whose feedback action was provided by the actuators, and the optimal gain matrices K , K_{det} , K_{slew} and K_{point} , each one associated to one phase, were computed through optimal control (*lqr* command of MATLAB). The quantity that is minimised by the optimal control is the following:

$$J_0 = \frac{1}{2} x_f^T S x_f + \frac{1}{2} \int_0^{t_f} x^T (Q + P^T B R^{-1} B^T P) x dt \quad (24)$$

And the control action is computed linearly:

$$M_c = -K \cdot x \quad (25)$$

Where x is the state associated to each phase.

In order to compute the optimal gains, 4 matrices, A, B, Q, and R had to be computed for each phase.

5.3.1 De-tumbling and switch to slew manoeuvre

The spacecraft is rotating with random angular velocities and the objective of the de-tumbling phase is to reduce them as much as possible, in order to allow sensors' initialization and to start the slew manoeuvre, once the on-board system computes its attitude. For this phase the state that needs to be minimized is the angular velocity estimated by the system, and the matrices are the following:

$$A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} \frac{1}{I_x} & 0 & 0 \\ 0 & \frac{1}{I_y} & 0 \\ 0 & 0 & \frac{1}{I_z} \end{bmatrix}$$

$$Q = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad R = \begin{bmatrix} \frac{1}{(10^{-1})^2} & 0 & 0 \\ 0 & \frac{1}{(10^{-1})^2} & 0 \\ 0 & 0 & \frac{1}{(10^{-1})^2} \end{bmatrix}$$

Matrices A and B derive from a linearized state space form, while the gain matrix Q is simply an identity matrix, since the angular velocity needs to be minimized, while R, associated to the control effort was found through a trial and error process aimed to avoid obtaining excessive control moments.

Once the norm of the estimated angular velocities falls below a threshold (0.0008 rad/s), the slew manoeuvre starts.

5.3.2 Slew manoeuvre and switch to pointing

This mission phase aims to rotate the spacecraft towards the desired direction, while reducing as much as possible the angular velocity. The mission's objective is to perform Nadir pointing, which translates into making $A_{B/L}$ as close as possible to an identity matrix. As a consequence, the state for this phase was formed by the estimated angular velocity and the means, with sign change, of the extra-diagonal elements of the estimated $A_{B/L}$ matrix. Here are shown the four matrices related to the optimal control for the slew manoeuvre:

$$A = \begin{bmatrix} 0 & (1 - K_y)n & 0 & K_y n^2 & 0 & 0 \\ (K_r - 1)n & 0 & 0 & 0 & -4K_r n^2 & 0 \\ 0 & 0 & 0 & -3K_p n^2 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} \frac{1}{I_x} & 0 & 0 \\ 0 & \frac{1}{I_y} & 0 \\ 0 & 0 & \frac{1}{I_z} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$Q = \begin{bmatrix} \frac{1}{(5 \cdot 10^{-3})^2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{(5 \cdot 10^{-3})^2} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{(5 \cdot 10^{-3})^2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{(5 \cdot 10^{-3})^2} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{(5 \cdot 10^{-3})^2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{(5 \cdot 10^{-3})^2} \end{bmatrix}, \quad R = \begin{bmatrix} \frac{1}{(10^{-3})^2} & 0 & 0 \\ 0 & \frac{1}{(10^{-3})^2} & 0 \\ 0 & 0 & \frac{1}{(5 \cdot 10^{-3})^2} \end{bmatrix}$$

The first 2 matrices derive from linearized equations for the gravity gradient expressed in the LVLH frame, where K_y, K_r and K_p are the coefficients associated with yaw, roll and pitch axis, respectively, while n is the mean angular speed of the spacecraft on the orbit. More precisely, they would be associated to a linearized form of dynamic equations where the state would be formed by the angles between Body and LVLH frames and their derivatives, but in case of large angles and rotations, it is still possible to choose and compute the matrixes and the gains of the linearized case, and change only the state that multiplies the gain. In order to obtain the orders of magnitude for the values of the last matrices Q and R, these formulas were considered:

$$Q = \text{diag} \left(\frac{1}{x_{i\max}^2} \right) \quad R = \text{diag} \left(\frac{1}{u_{i\max}^2} \right)$$

To estimate $x_{i\max}$ and $u_{i\max}$ the accuracies of the sensors and the maximum torques that could be provided by the actuators were taken into account. Once the body frame is close enough to the LVLH one, which is translated into a condition on the maximum extra-diagonal element of $A_{B/L}$ estimated (0.005).

5.3.3 Nadir pointing

This is the final phase of the mission, where the aim is to perform Earth pointing. Similarly to the slew manoeuvre, the non diagonal elements of $A_{B/L}$ must get closer to zero, but in this case

we want the body frame to behave like LVLH RF, meaning that also the derivatives of the angles between the 2 reference frames need to be minimised. As a consequence, the first 3 elements of the state are those derivatives, computed as it follows, since the LVLH is rotating with angular velocity $\begin{bmatrix} 0 & 0 & n \end{bmatrix}^T$:

$$\begin{bmatrix} \dot{\alpha}_x \\ \dot{\alpha}_y \\ \dot{\alpha}_z \end{bmatrix} = \begin{bmatrix} w_x \\ w_y \\ w_z \end{bmatrix} - A_{B/L} \begin{bmatrix} 0 \\ 0 \\ n \end{bmatrix}$$

The matrices associated with this phase are exactly the same as the ones adopted for the previous one, since the maximum effort done by the actuators doesn't change, nor do the sensors' accuracies, but in this case the state is exactly the one expected for that formulation.

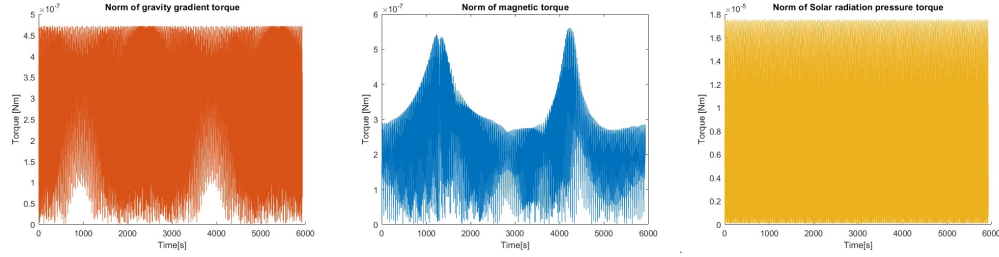
Note: in order to compute the control performances on board, and some elements of the state, the matrix $A_{N/L}$ had to be known. Since $A_{B/N}$ can be estimated, but the real $A_{N/L}$ is not available on board, we assumed that a propagation of the true anomaly θ could be performed on board, leading to the knowledge of $A_{N/L} = A_{L/N}^T$

$$A_{L/N} = \begin{bmatrix} \cos(\theta) & \sin(\theta) & 0 \\ -\sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(i) & \sin(i) \\ 0 & -\sin(i) & \cos(i) \end{bmatrix} \quad (26)$$

6 Final results

6.1 Perturbations

Here is presented the plotted showing the evolution of the disturbing torques during 1 revolution of the spacecraft on its orbit:



(a) Norm of the gravity gradient torque (b) Norm of the magnetic torque (c) Norm of the solar radiation pressure torque

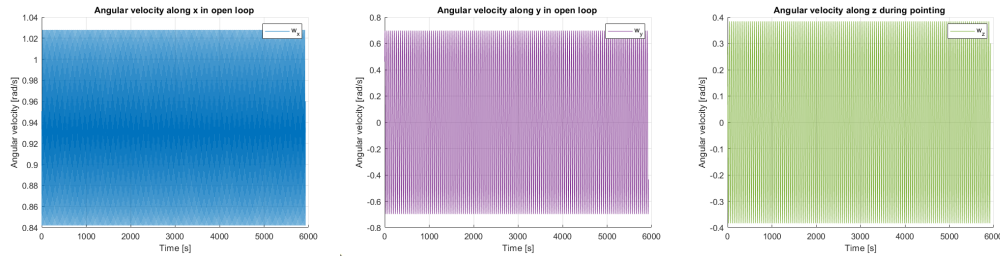
6.2 Initial conditions

The initial attitude is completely random and was built using the Rodrigues formula, with a rotation of a random angle around a random axis.

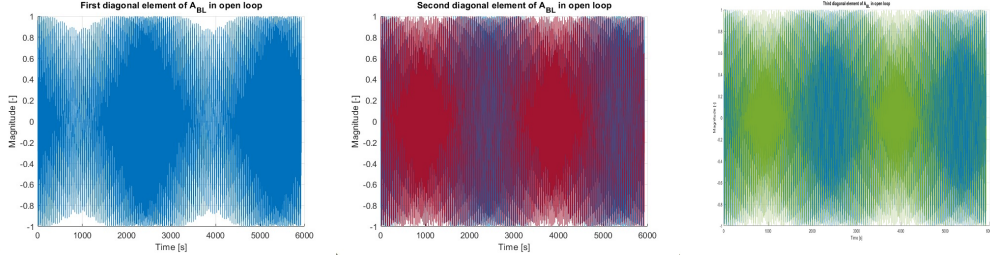
Similarly, the initial angular velocities are random numbers in the range $[0,1] \frac{rad}{s}$

6.3 Open-loop system

The behaviour of the open loop system, which consists of environment, dynamics and kinematics blocks, is shown below:



(d) Angular velocity along x axis in open loop (e) Angular velocity along y axis in open loop (f) Angular velocity along z axis in open loop



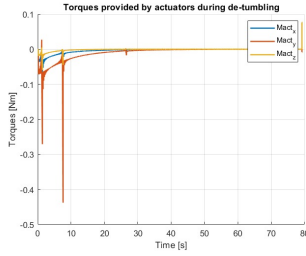
(g) 1st diagonal element of A_{BL} in open loop (h) 2nd diagonal element of A_{BL} in open loop (i) 3rd diagonal element of A_{BL} in open loop

6.4 Closed-loop system

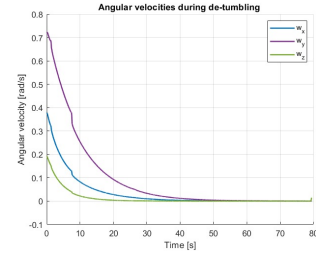
In this section all the main parameters of each mission phase will be reported.

6.4.1 De-tumbling

Here the most important parameter is the angular velocity, that must be lowered as much as possible. This phase starts at $t = 0s$, and ends when $\|w_{estimated}\| \leq 0.005 rad/S$, which happens around $t = 80s$

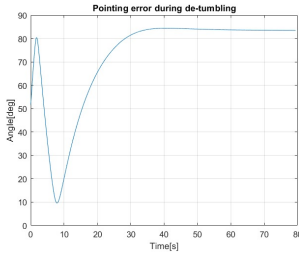


(j) Actuators' torques during de-tumbling

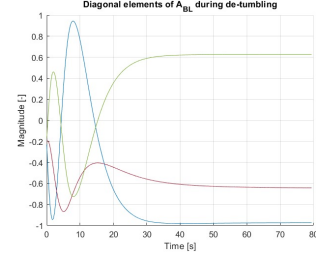


(k) Angular velocities during de-tumbling

Figure 3: De-tumbling actuator torques and angular velocities

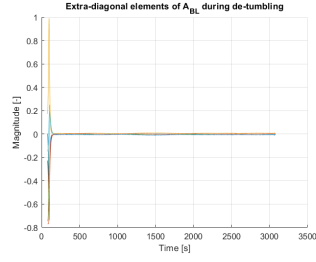


(a) Pointing error during de-tumbling



(b) Diagonal elements of A_{BL} during de-tumbling

Figure 4: Pointing error and diagonal elements of A_{BL} during de-tumbling



(a) Extra-diagonal elements of A_{BL} during de-tumbling

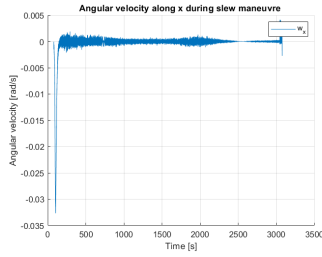
Figure 5: Extra-diagonal elements of A_{BL} during de-tumbling

6.4.2 Slew Manoeuvre

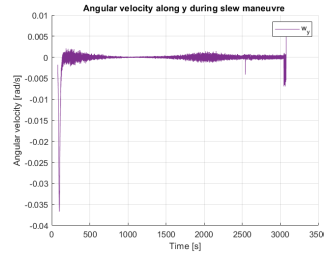
This phase starts as soon as the de-tumbling ends (around $t = 80s$) and goes on until

$$\max(A_{B/L,ij}) \leq 0.005 \quad \text{with } i \neq j$$

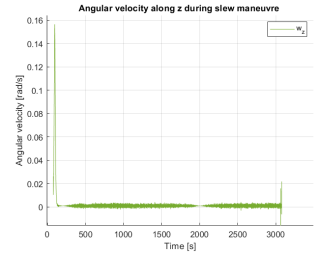
This condition is verified at time instants around 3000s. In the formula above the matrix considered is $A_{B/L\text{estimated}}$, obtained as $A_{B/N\text{estimated}} \cdot A_{N/L}$ (angular velocity, pointing error, torque act and AB/L)



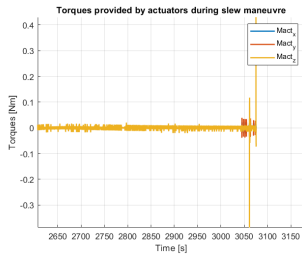
(a) Angular velocity along x axis during slew manoeuvre



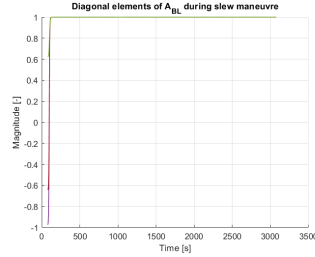
(b) Angular velocity along y axis during slew manoeuvre



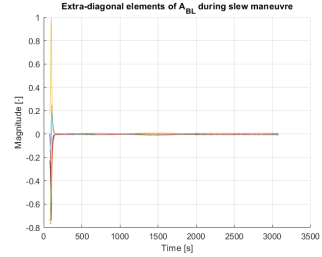
(c) Angular velocity along z axis during slew manoeuvre



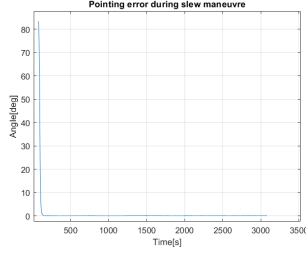
(d) Actuators' torques during slew manoeuvre



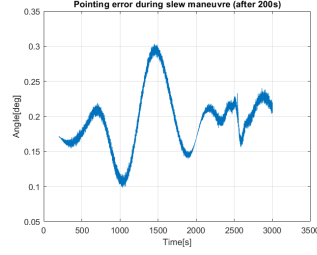
(e) Diagonal elements of A_{BL} during slew manoeuvre



(f) Extra-diagonal elements of A_{BL} during slew manoeuvre



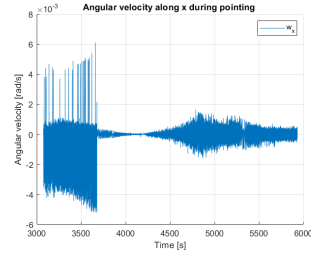
(g) Pointing error during slew manoeuvre



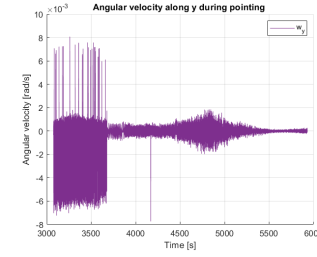
(h) Pointing error detail during slew manoeuvre

6.4.3 Earth pointing

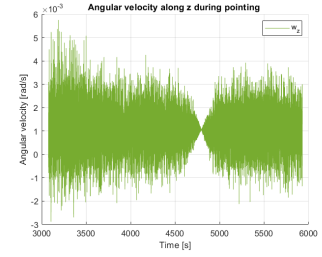
When $A_{B/Nestimated}$ and $A_{L/N}$ are similar enough ($t \approx 3000s$) the pointing phase begins, leading to the following results: (angular velocity, pointing error, torqueact and AB/L)



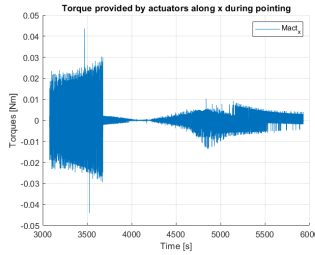
(i) Angular velocity along x axis during Earth pointing



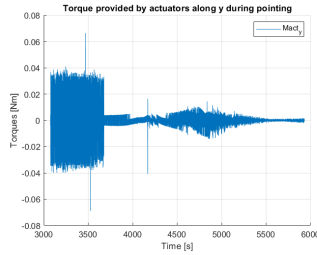
(j) Angular velocity along y axis during Earth pointing



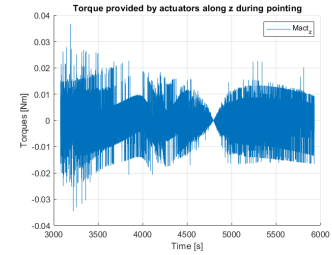
(k) Angular velocity along z axis during Earth pointing



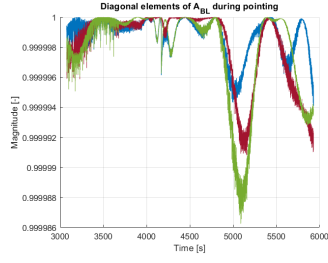
(l) Actuators' torque along x during Earth pointing



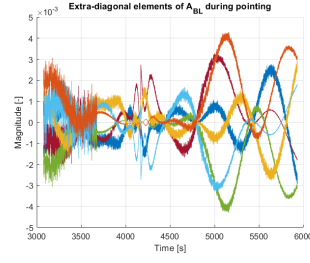
(m) Actuators' torque along y during Earth pointing



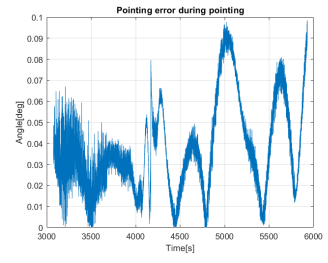
(n) Actuators' torque along z during Earth pointing



(o) Actuators' torque along x during Earth pointing



(p) Actuators' torque along y during Earth pointing



(q) Actuators' torque along z during Earth pointing

6.5 Conclusions

Our objective was to maintain the pointing error below 0.5 degrees, and the graphs above demonstrate that this goal was successfully achieved.

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URL: <https://agupubs.onlinelibrary.wiley.com/doi/full/10.1029/2023EA003035>